Decomposition of Concentration-Index Trees

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1 4. Results

The results section reports only the objects needed to answer the main technical question: whether inequality-oriented recursive partitioning identifies subgroup structure that is visible, stable, and comparable with classical decomposition output. Full generated outputs are moved to the appendix.

1.1 4.1 Exploratory Data Analysis

Figure 1: Under-five mortality distribution and DHS wealth index in the DRC analysis data.

1.2 4.2 Application of developed tree models to the DRC DHS Data

1.2.1 4.2.1 Comparison of Indirect Decomposition Results

Categorical predictors enter the regression decomposition as separate one-hot terms. The predictive ranger is fitted on the same design-matrix terms, so the comparison table reports categorical levels at the same granularity instead of collapsing them back to their parent variable.

Table 1: Regression and predictive-forest SHAP percentage contributions by criterion

	Regression	Predictive-forest SHAP

Variable	CI (%)	CIg (%)	CIc (%)	CI (%)	CIg (%)	CIc (%)	L (%)
Residence	33.993	70.954	70.954	25.894	25.894	25.894	27.465
Province: Lualaba	6.713	1.192	1.192	-1.214	-1.214	-1.214	-1.402
Province: Sankuru	6.700	1.570	1.570	0.928	0.928	0.928	0.194
Partner occupation: d Agriculture	6.406	7.369	7.369	7.496	7.496	7.496	7.252
Province: Sud-Kivu	4.471	0.665	0.665	4.584	4.584	4.584	5.553
Birth group: c 2-4 long	3.506	3.503	3.503	0.738	0.738	0.738	3.313
Mother occupation: d Agriculture	3.463	4.521	4.521	10.189	10.189	10.189	9.494
Province: Kwilu	3.342	0.326	0.326	-0.719	-0.719	-0.719	-0.518
Province: Nord-Kivu	2.859	0.243	0.243	-0.884	-0.884	-0.884	-0.091
Partner education	2.750	0.684	0.684	3.087	3.087	3.087	2.224

1.2.2 4.2.2 Root-Node Inequality

Table 2: Root-node inequality in under-five mortality by concentration-index criterion

Root node	CI	CIg	CIc	L
Full sample	0.1112	0.0077	0.0308	0.0818

1.2.3 4.2.3 Selected Concentration-Index Tree

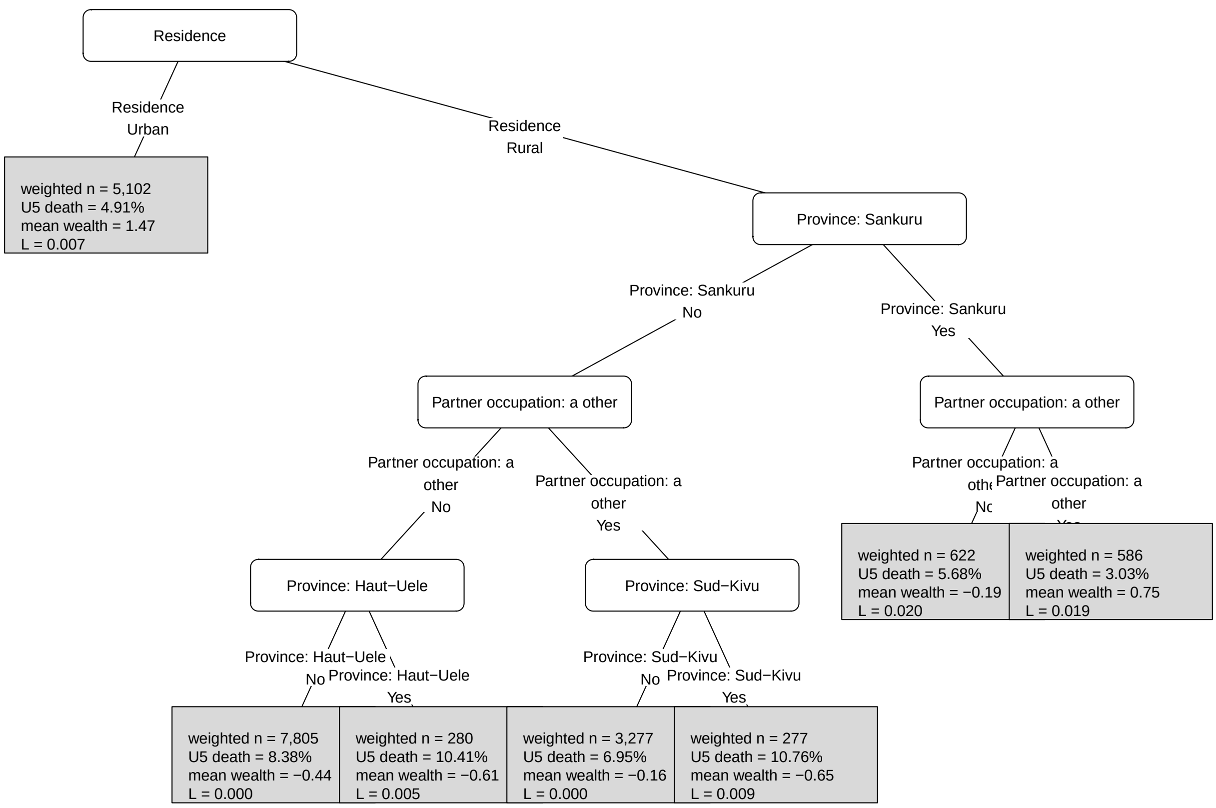

Figure 2: Selected concentration-index tree for under-five mortality in the DRC.

1.2.4 Surrogate Tree Comparison

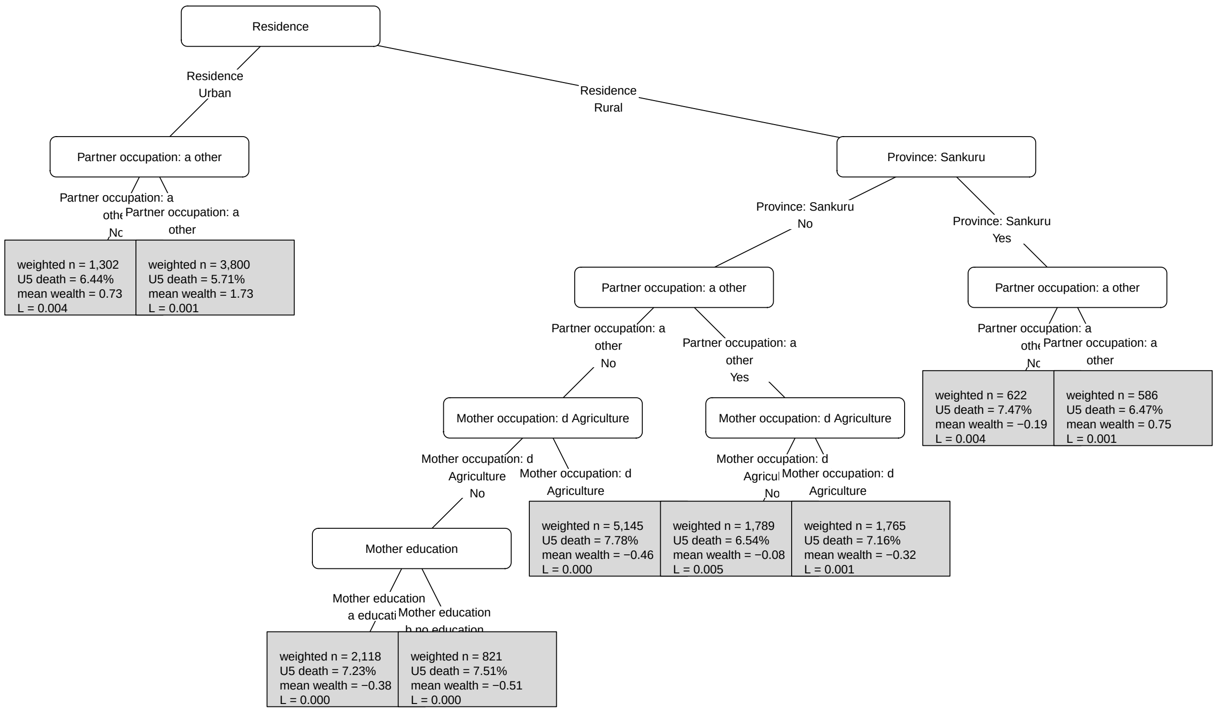

Figure 3: Surrogate tree summarising the selected concentration-index forest predictions.

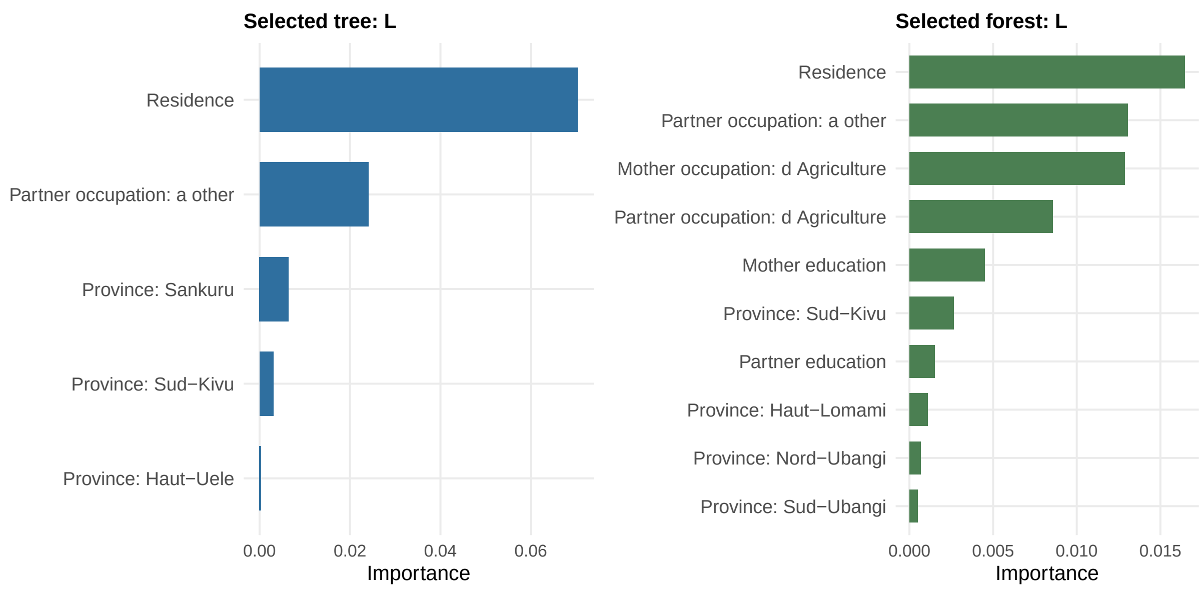

Figure 4: Variable importance for the selected concentration-index tree and forest.

1.3 4.3 Simulation Study

1.3.1 4.3.1 Recovery of the Simulated Subgroup Structure

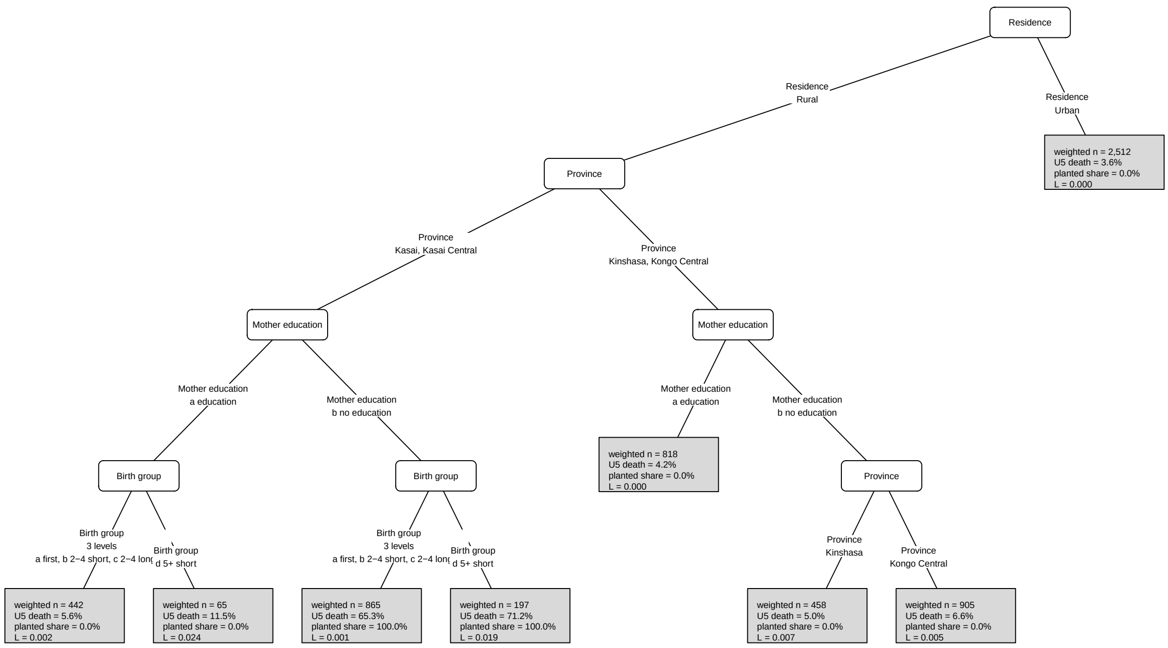

Figure 5: Tree recovered from simulated data with a known subgroup-generating mechanism.

2 Appendix

2.1 Results

2.1.1 Appendix R1: Exploratory Data Outputs

Table 3: Weighted distribution of analysis variables by under-five mortality status

	Under-five mortality		Overall
Variable	**Alive at age 5**	**Died before age 5**	**Overall**
	(weighted N=16,704)	(weighted N=1,244)	(weighted N=17,948)
Low-skill HH			
No	10,185.6 (61.0%)	786.1 (63.2%)	10,971.7 (61.1%)
Yes	6,518.4 (39.0%)	458.2 (36.8%)	6,976.5 (38.9%)
Child sex			
Female	8,099.9 (48.5%)	515.2 (41.4%)	8,615.1 (48.0%)
Male	8,604.1 (51.5%)	729 (58.6%)	9,333.1 (52.0%)
Birth group			
a first	2,691.8 (16.1%)	242.1 (19.5%)	2,933.9 (16.3%)
b 2-4 short	2,112.9 (12.6%)	186.1 (15.0%)	2,299 (12.8%)
c 2-4 long	5,861.4 (35.1%)	287 (23.1%)	6,148.4 (34.3%)
d 5+ short	1,633.8 (9.8%)	248.3 (20.0%)	1,882.1 (10.5%)
e 5+ long	4,404.1 (26.4%)	280.7 (22.6%)	4,684.8 (26.1%)

Mother age			
a20 or more	16,100.6 (96.4%)	1,172.9 (94.3%)	17,273.4 (96.2%)
less than 20	603.4 (3.6%)	71.4 (5.7%)	674.8 (3.8%)
Residence			
Urban	4,851.8 (29.0%)	250.3 (20.1%)	5,102.1 (28.4%)
Rural	11,852.2 (71.0%)	993.9 (79.9%)	12,846.1 (71.6%)
Mother education			
a education	13,472.5 (80.7%)	958 (77.0%)	14,430.5 (80.4%)
b no education	3,231.5 (19.3%)	286.2 (23.0%)	3,517.7 (19.6%)
Partner education			
a education	15,311.1 (91.7%)	1,106.1 (88.9%)	16,417.2 (91.5%)
b no education	1,392.9 (8.3%)	138.2 (11.1%)	1,531 (8.5%)
Mother occupation			
a other	3,999.5 (23.9%)	254.8 (20.5%)	4,254.4 (23.7%)
c Household, unskilled manual, not working	5,307.9 (31.8%)	350 (28.1%)	5,657.9 (31.5%)
d Agriculture	7,396.6 (44.3%)	639.4 (51.4%)	8,036 (44.8%)
Partner occupation			
a other	7,484.6 (44.8%)	455.9 (36.6%)	7,940.5 (44.2%)
c Household, unskilled manual, not working	2,704.3 (16.2%)	223 (17.9%)	2,927.3 (16.3%)
d Agriculture	6,515.1 (39.0%)	565.3 (45.4%)	7,080.4 (39.4%)

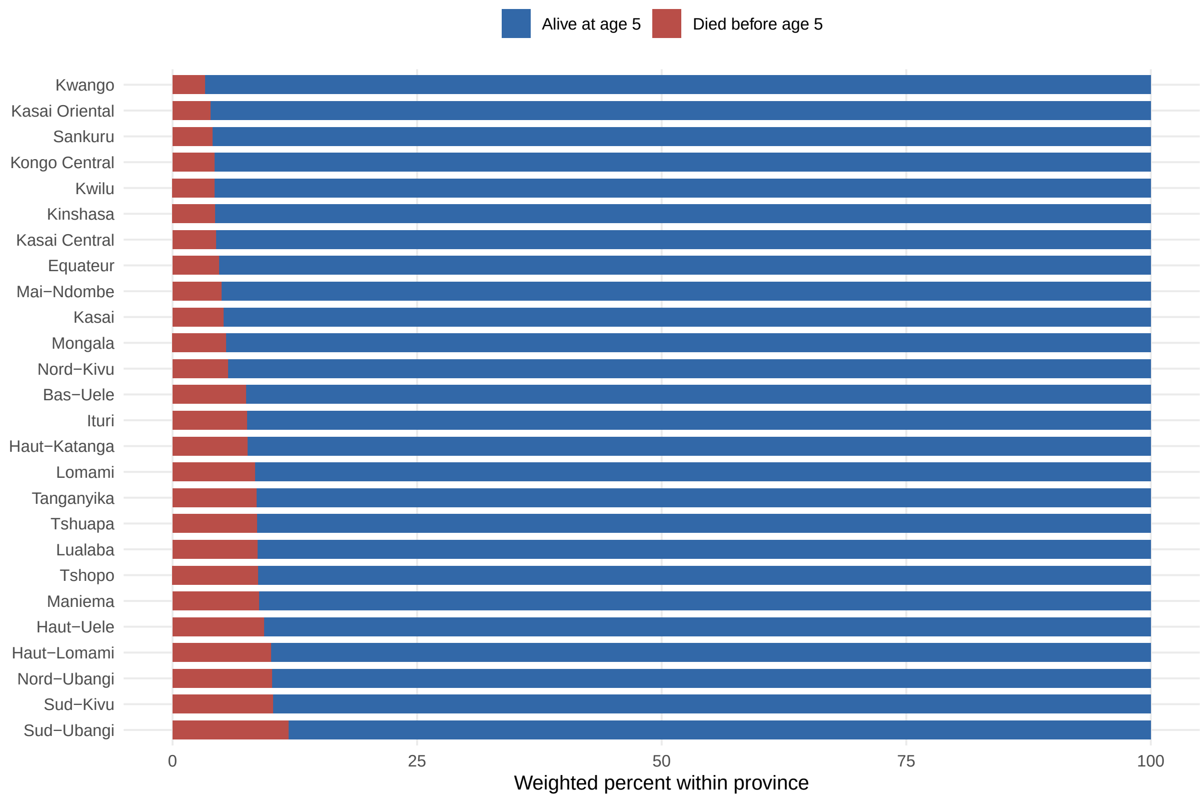

Figure 6: Weighted province distribution by under-five mortality status.

2.1.2 Appendix R2: Regression and Predictive-Forest Details

Regression model summary

Table 4: Survey-weighted regression model summary

Term	Estimate	Std. error	Statistic	p-value	Conf. low	Conf. high
Intercept	-2.983	0.268	-11.135	< 0.001	-3.509	-2.458
Low-skill HH	0.009	0.241	0.037	0.97074	-0.463	0.480
Child sex	0.306	0.085	3.591	< 0.001	0.139	0.472
Birth group: b 2-4 short	-0.003	0.144	-0.023	0.98128	-0.285	0.278
Birth group: c 2-4 long	-0.553	0.133	-4.153	< 0.001	-0.814	-0.292
Birth group: d 5+ short	0.434	0.146	2.980	0.00289	0.149	0.720
Birth group: e 5+ long	-0.372	0.136	-2.726	0.00642	-0.639	-0.104
Mother age	0.319	0.178	1.787	0.07393	-0.031	0.668
Residence	0.384	0.137	2.799	0.00514	0.115	0.654
Mother education	0.039	0.113	0.345	0.73043	-0.183	0.261
Partner education	0.199	0.151	1.319	0.18708	-0.097	0.496
Mother occupation: c Household, unskilled manual, not working	-0.087	0.223	-0.390	0.69656	-0.523	0.349
Mother occupation: d Agriculture	0.051	0.131	0.391	0.69544	-0.206	0.309
Partner occupation: c Household, unskilled manual, not working	0.151	0.177	0.854	0.39300	-0.196	0.498
Partner occupation: d Agriculture	0.103	0.105	0.980	0.32707	-0.103	0.309
Province: Kwango	-0.707	0.420	-1.684	0.09218	-1.530	0.116
Province: Kwilu	-0.540	0.317	-1.702	0.08886	-1.162	0.082
Province: Mai-Ndombe	-0.335	0.313	-1.070	0.28473	-0.949	0.279
Province: Kongo Central	-0.420	0.454	-0.926	0.35445	-1.310	0.469
Province: Equateur	-0.326	0.336	-0.972	0.33098	-0.984	0.331
Province: Mongala	-0.273	0.312	-0.877	0.38059	-0.884	0.338
Province: Nord-Ubangi	0.302	0.302	0.999	0.31797	-0.291	0.894
Province: Sud-Ubangi	0.489	0.336	1.457	0.14506	-0.169	1.147
Province: Tshuapa	0.176	0.321	0.549	0.58318	-0.453	0.806
Province: Kasai	-0.303	0.335	-0.906	0.36519	-0.960	0.353
Province: Kasai Central	-0.455	0.397	-1.146	0.25179	-1.232	0.323
Province: Kasai Oriental	-0.615	0.606	-1.016	0.30961	-1.803	0.572
Province: Lomami	0.133	0.321	0.414	0.67890	-0.496	0.761
Province: Sankuru	-0.646	0.335	-1.930	0.05366	-1.302	0.010
Province: Haut-Katanga	-0.002	0.315	-0.007	0.99429	-0.619	0.615
Province: Haut-Lomami	0.368	0.285	1.288	0.19765	-0.192	0.927
Province: Lualaba	0.377	0.297	1.267	0.20508	-0.206	0.959
Province: Tanganyika	0.154	0.292	0.528	0.59752	-0.419	0.727
Province: Maniema	0.172	0.334	0.516	0.60586	-0.482	0.826
Province: Nord-Kivu	-0.368	0.347	-1.058	0.29019	-1.049	0.314
Province: Bas-Uele	0.137	0.291	0.470	0.63804	-0.433	0.707
Province: Haut-Uele	0.197	0.323	0.611	0.54116	-0.436	0.831
Province: Ituri	0.232	0.311	0.746	0.45549	-0.378	0.842
Province: Tshopo	0.186	0.328	0.566	0.57124	-0.457	0.829
Province: Sud-Kivu	0.335	0.292	1.146	0.25199	-0.238	0.908

Predictive ranger tuning results

Table 5: Weighted class distribution before and after ranger undersampling

Sample	Outcome	Rows	Weighted N	Weighted percent

Before undersampling	Alive at age 5	18589	16703.99	93.07
Before undersampling	Died before age 5	1454	1244.24	6.93
After undersampling	Alive at age 5	5681	4977.15	80.00
After undersampling	Died before age 5	1454	1244.24	20.00

Table 6: Cross-validation results for the predictive ranger model

Metric	mtry	min_n	Mean	Std. error	Folds	Rank
accuracy	1	10	0.7962	0.0001	5	1
accuracy	1	80	0.7962	0.0001	5	2
accuracy	1	150	0.7962	0.0001	5	3
accuracy	15	150	0.7943	0.0011	5	4
accuracy	29	150	0.7940	0.0012	5	5
accuracy	43	150	0.7931	0.0014	5	6
accuracy	15	80	0.7924	0.0013	5	7
accuracy	43	80	0.7899	0.0014	5	8
accuracy	29	80	0.7898	0.0014	5	9
accuracy	15	10	0.7741	0.0014	5	10
accuracy	29	10	0.7678	0.0013	5	11
accuracy	43	10	0.7658	0.0011	5	12
roc_auc	1	80	0.6270	0.0108	5	1
roc_auc	1	10	0.6262	0.0117	5	2
roc_auc	1	150	0.6242	0.0112	5	3
roc_auc	15	150	0.6113	0.0099	5	4
roc_auc	29	150	0.6080	0.0098	5	5
roc_auc	43	150	0.6064	0.0103	5	6
roc_auc	15	80	0.6056	0.0114	5	7
roc_auc	29	80	0.6039	0.0117	5	8
roc_auc	43	80	0.6037	0.0114	5	9
roc_auc	15	10	0.5871	0.0145	5	10
roc_auc	29	10	0.5823	0.0141	5	11
roc_auc	43	10	0.5799	0.0145	5	12
sensitivity	43	10	0.1162	0.0094	5	1
sensitivity	29	10	0.1094	0.0069	5	2
sensitivity	15	10	0.0936	0.0091	5	3
sensitivity	43	80	0.0289	0.0058	5	4
sensitivity	29	80	0.0255	0.0068	5	5
sensitivity	15	80	0.0179	0.0048	5	6
sensitivity	43	150	0.0151	0.0037	5	7
sensitivity	29	150	0.0124	0.0034	5	8
sensitivity	15	150	0.0048	0.0014	5	9
sensitivity	1	10	0.0000	0.0000	5	10
sensitivity	1	80	0.0000	0.0000	5	11
sensitivity	1	150	0.0000	0.0000	5	12

Predictive ranger SHAP summary plot

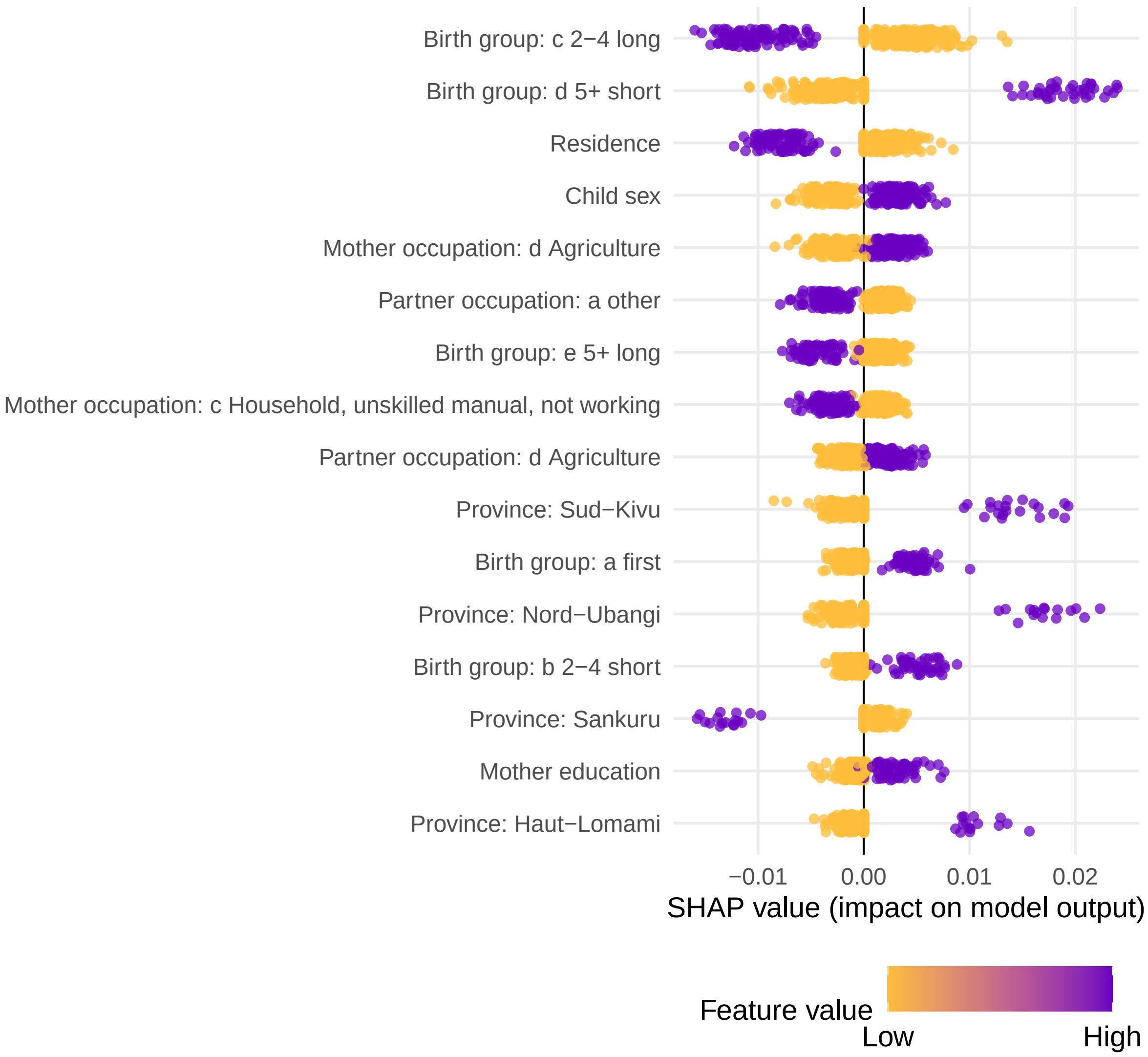

Figure 7: SHAP summary plot for the predictive ranger model.

2.1.3 Appendix R3: Concentration-Index Tree Details

Tree tuning and selection details

Table 7: Best tree hyperparameters selected by validation gain within each impurity criterion

type	minsplit	minbucket	minprob	maxdepth	min_gain	min_relative_gain
CI	500	250	0.01	10	0	0.2
CIg	500	250	0.01	10	0	0.2
CIc	500	250	0.01	10	0	0.2
L	500	250	0.01	10	0	0.2

Tree fit and selection summaries

Table 8: Tree summary for the selected settings

criterion	metric	estimate	std__error
CI	Root impurity	0.1115	NA
CI	Training gain	0.0505	0.0030
CI	Training gain (% root)	45.5193	2.7884
CI	Validation gain	0.0067	0.0122
CI	Validation gain (% root)	-9.2483	14.5979
CIc	Root impurity	0.0306	NA
CIc	Training gain	0.0190	0.0008
CIc	Training gain (% root)	61.6639	2.4017
CIc	Validation gain	-0.0003	0.0062
CIc	Validation gain (% root)	-37.2283	30.9978
CIg	Root impurity	0.0076	NA
CIg	Training gain	0.0048	0.0002
CIg	Training gain (% root)	61.6639	2.4017
CIg	Validation gain	-0.0001	0.0015
CIg	Validation gain (% root)	-37.2283	30.9978
L	Root impurity	0.0878	NA
L	Training gain	0.0775	0.0013
L	Training gain (% root)	94.7077	0.3721
L	Validation gain	0.0641	0.0184
L	Validation gain (% root)	64.4396	9.8198

Tree selection metrics

Table 9: Selection metrics for fitted concentration-index trees

Criterion	Root objective	Train gain	Train gain/root	Validation gain	Validation gain/root	Terminal nodes
CI	0.1115	0.0505	0.4552	0.0067	-0.0925	3.9
CIc	0.0306	0.0190	0.6166	-0.0003	-0.3723	6.1
CIg	0.0076	0.0048	0.6166	-0.0001	-0.3723	6.1
L	0.0878	0.0775	0.9471	0.0641	0.6444	6.0

Tree validation plots

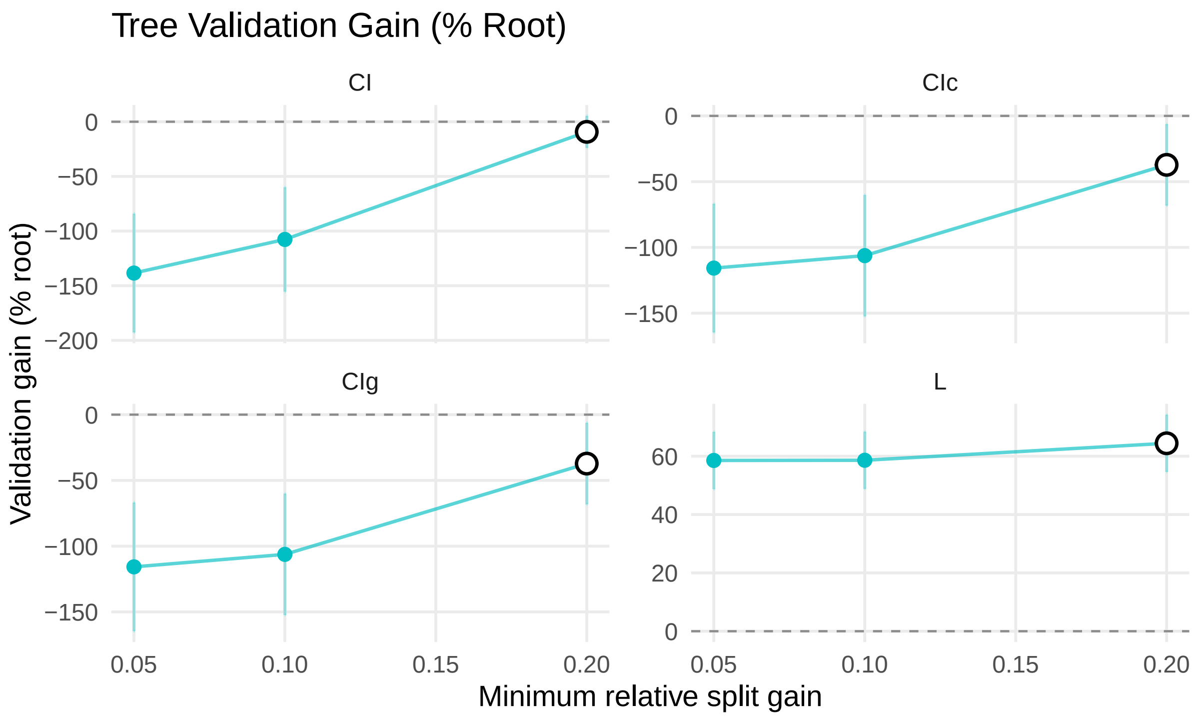

Figure 8: Tree validation gain and complexity by minimum relative split gain.

Tree complexity plots

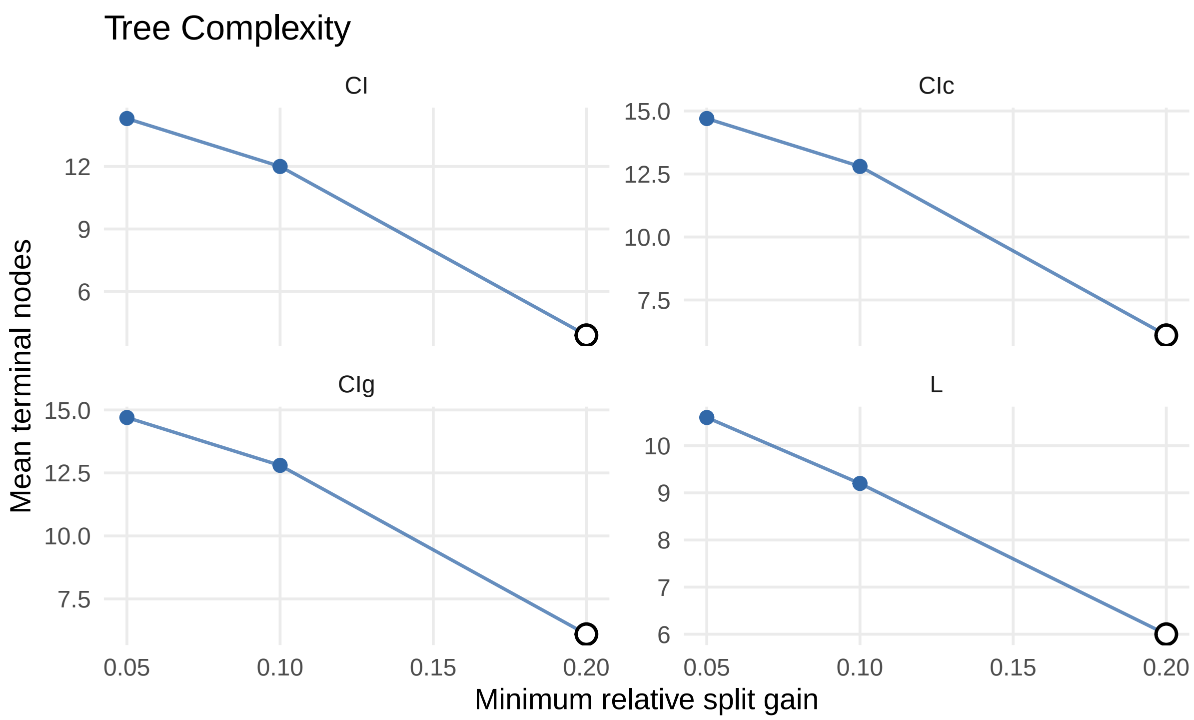

Figure 9: Tree complexity by number of terminal nodes.

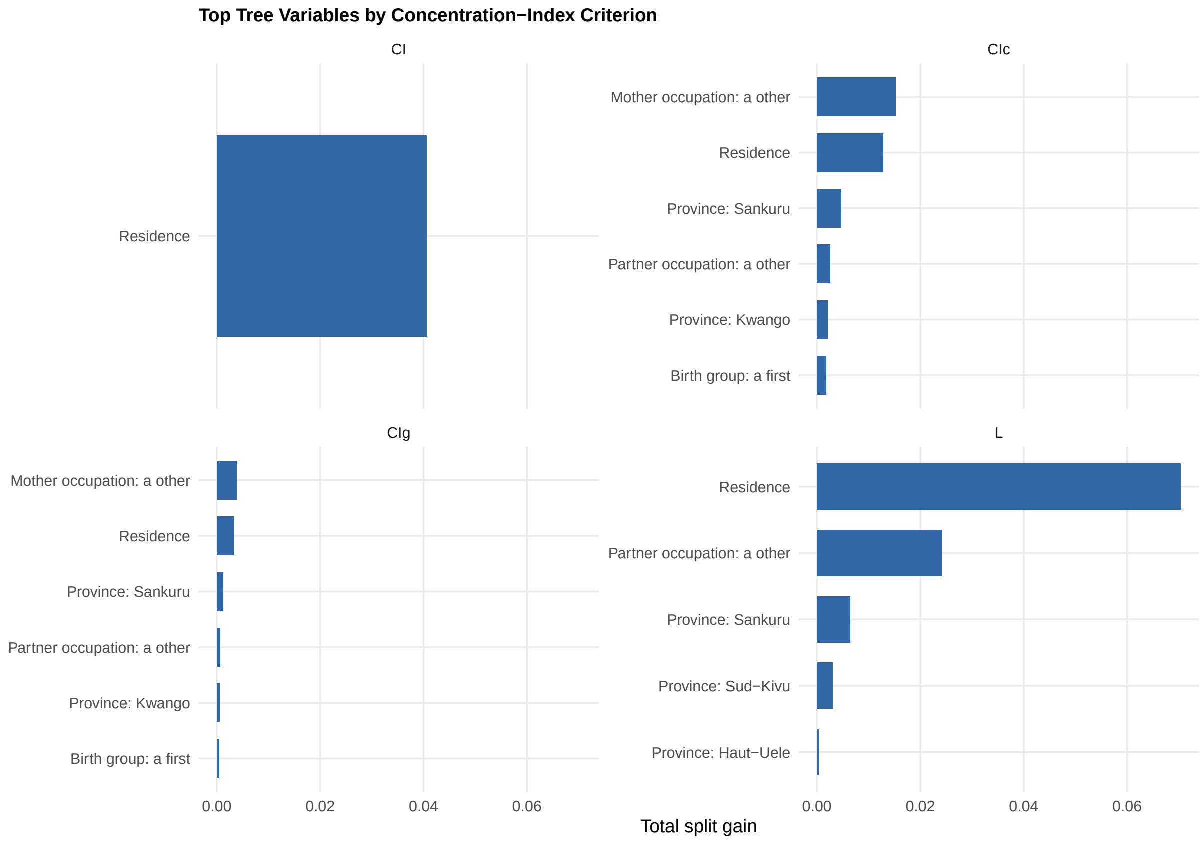

Figure 10: Top tree variables by concentration-index criterion.

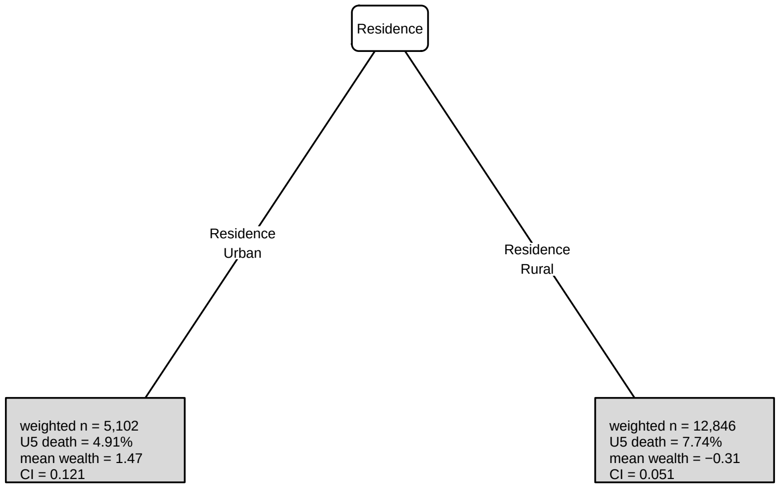

Figure 11: Concentration-index tree fitted with the CI criterion.

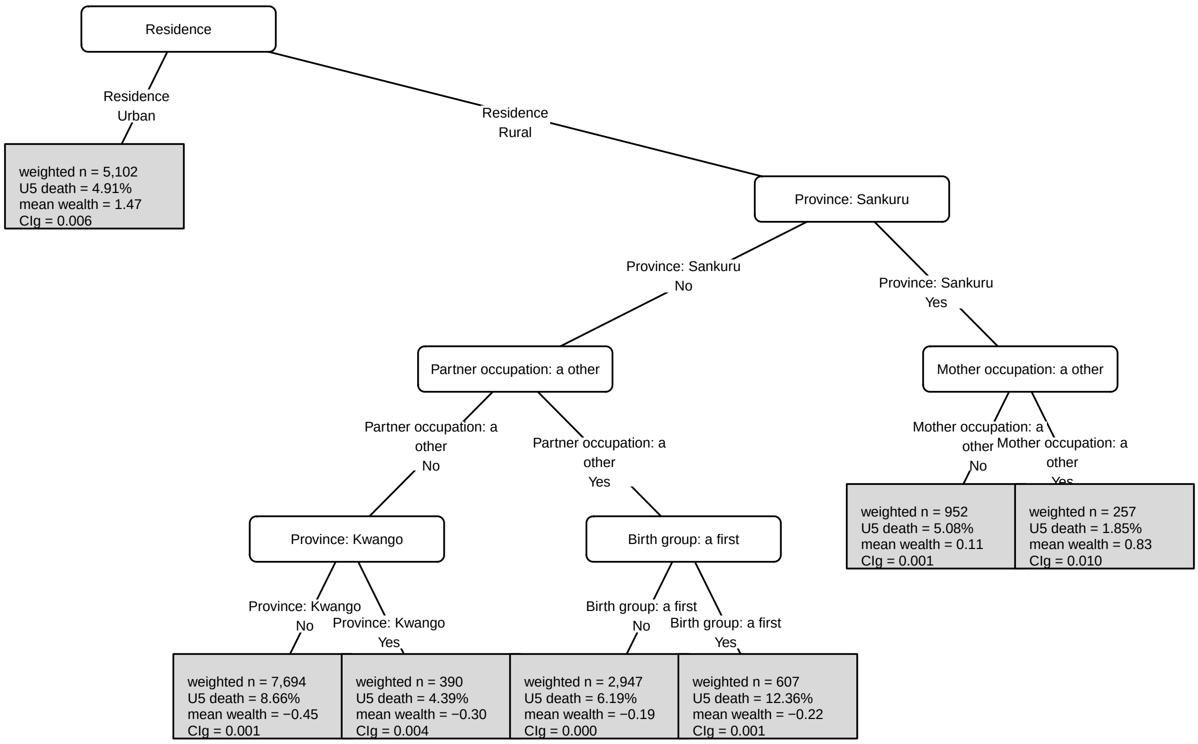

Figure 12: Concentration-index tree fitted with the CIg criterion.

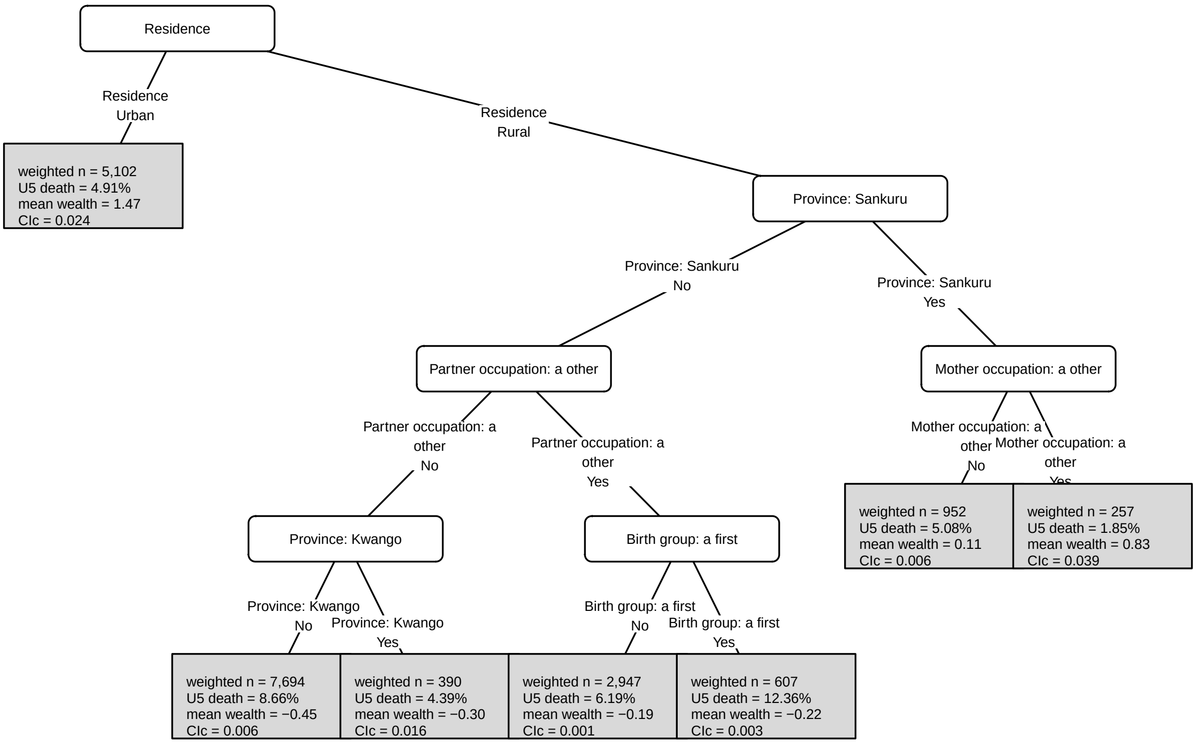

Figure 13: Concentration-index tree fitted with the CIc criterion.

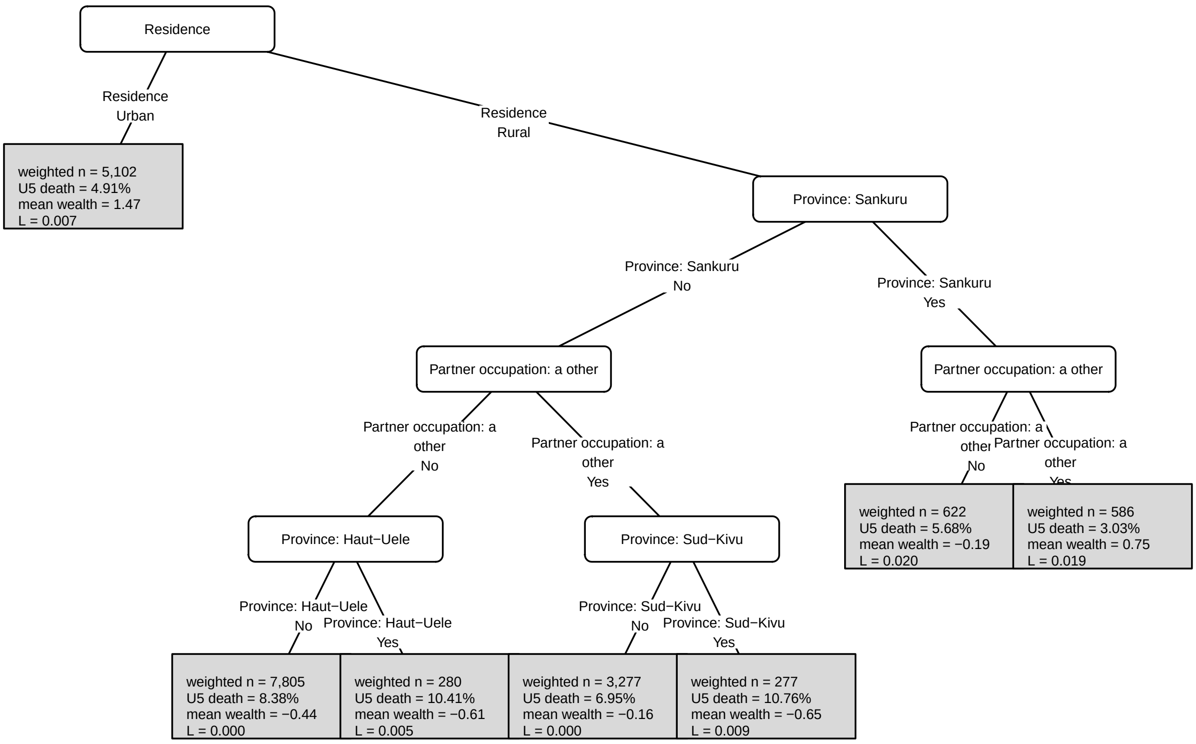

Figure 14: Concentration-index tree fitted with the L criterion.

2.1.4 Appendix R4: Concentration-Index Forest Details

Forest tuning and selection details

Table 10: Best forest hyperparameters selected by validation gain within each impurity criterion

type	ntree	mtry	minsplit	minbucket	minprob	maxdepth	min_gain	min_relative_gain
CI	200	6	500	250	0.01	10	0	0.20
CIc	200	6	500	250	0.01	10	0	0.20
CIg	200	6	500	250	0.01	10	0	0.20
L	200	6	500	250	0.01	10	0	0.05

Forest fit and selection summaries

Table 11: Forest summary for the selected settings

criterion	metric	estimate	std_error
CI	Root impurity	0.1115	NA
CI	Training gain	0.0079	0.0009
CI	Training gain (% root)	7.1122	0.7003
CI	Validation gain	0.0005	0.0033
CI	Validation gain (% root)	-5.0667	5.7159
CIc	Root impurity	0.0306	NA
CIc	Training gain	0.0040	0.0003
CIc	Training gain (% root)	12.8252	0.7148
CIc	Validation gain	0.0003	0.0016
CIc	Validation gain (% root)	-8.9120	9.4671
CIg	Root impurity	0.0076	NA
CIg	Training gain	0.0009	0.0001
CIg	Training gain (% root)	12.1891	0.9564
CIg	Validation gain	0.0000	0.0004
CIg	Validation gain (% root)	-9.6012	9.8384
L	Root impurity	0.0878	NA
L	Training gain	0.0589	0.0014
L	Training gain (% root)	71.8363	0.4428
L	Validation gain	0.0513	0.0135
L	Validation gain (% root)	48.8379	7.4994

Forest validation plots

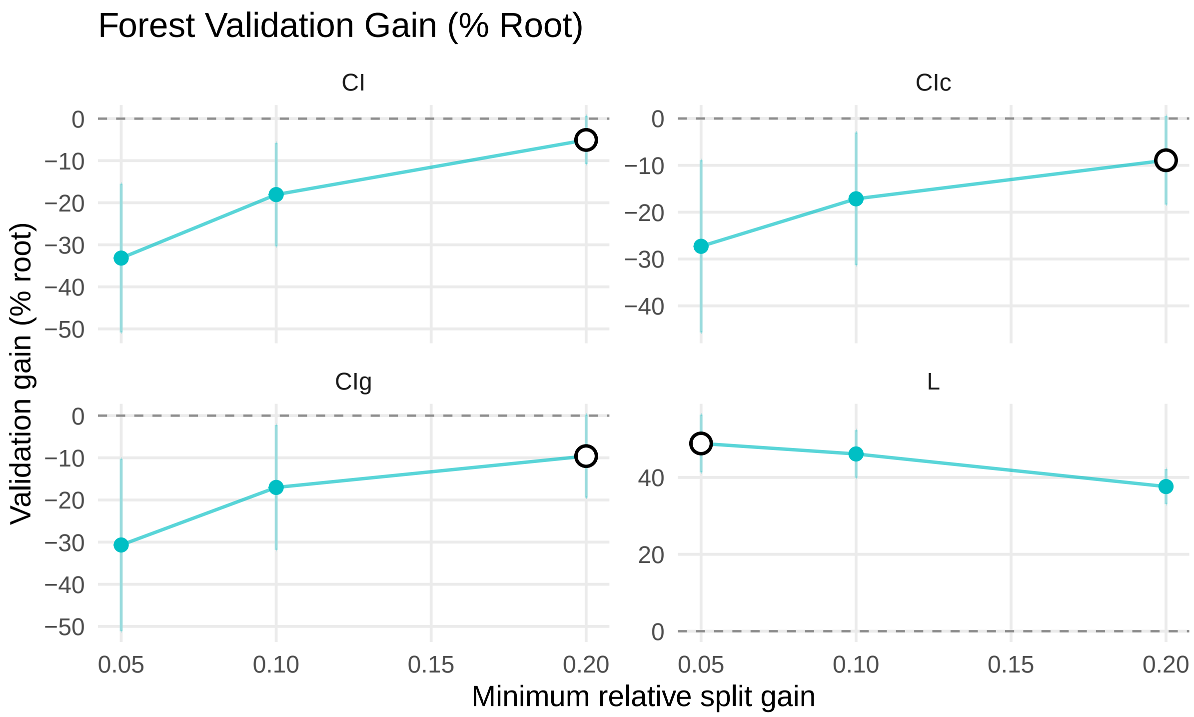

Figure 15: Forest validation gain and complexity by minimum relative split gain.

Forest complexity plots

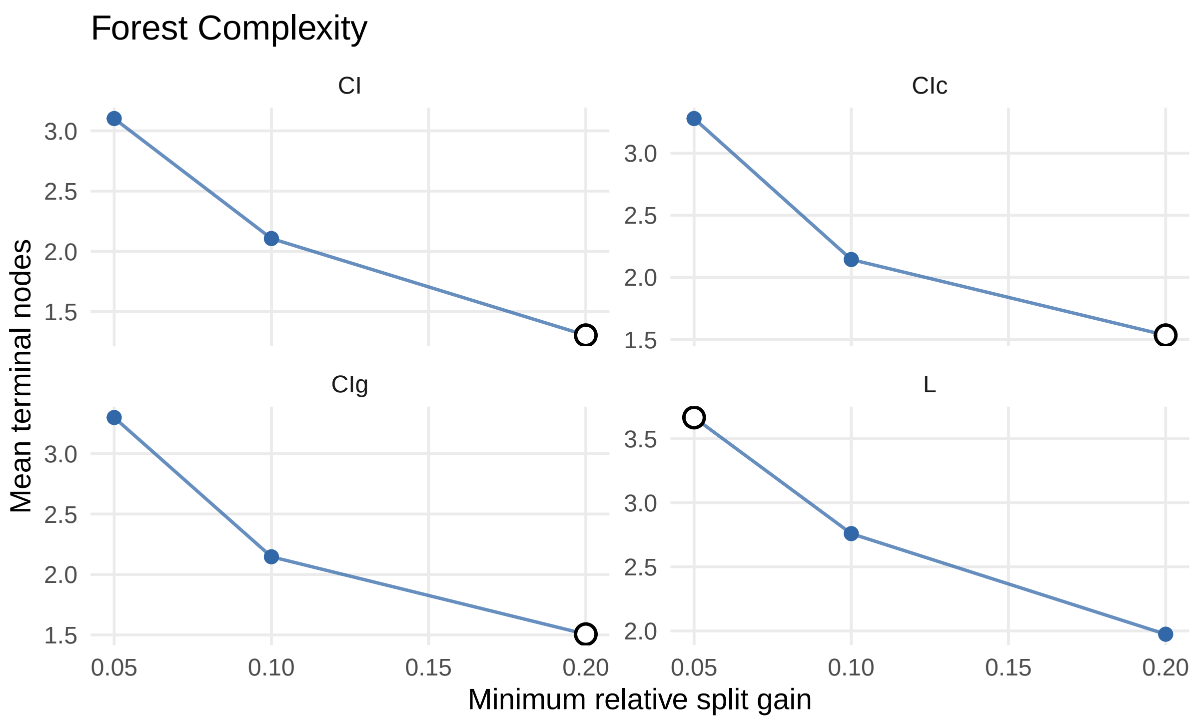

Figure 16: Forest complexity by number of terminal nodes.

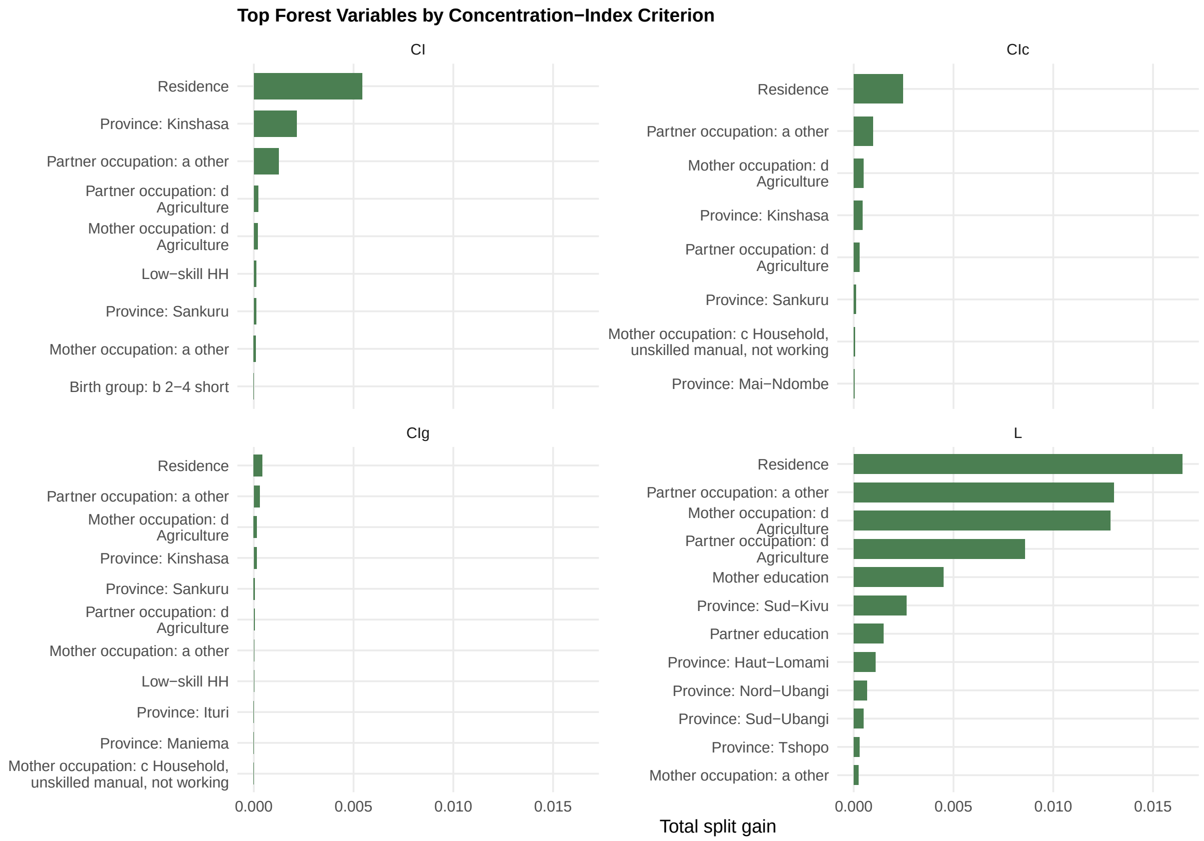

Figure 17: Top forest variables by concentration-index criterion.

Non-Selected Forest Surrogate Trees

The selected forest surrogate appears in the results section. The remaining criterion-specific forest surrogates are retained here for comparison.

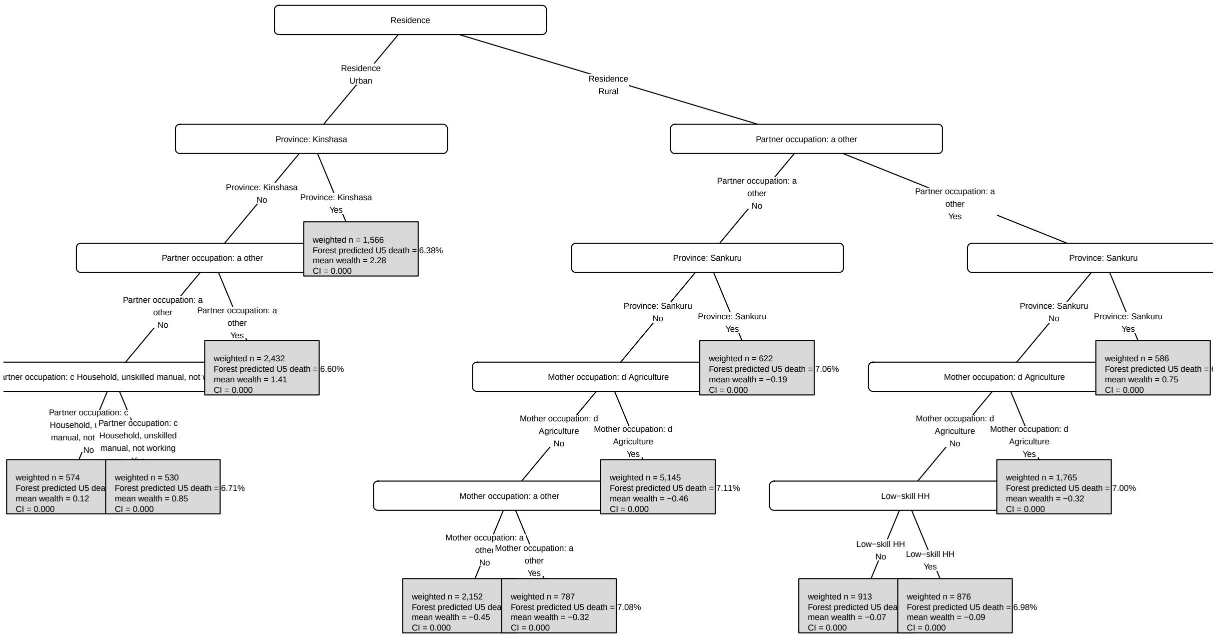

Figure 18: Surrogate tree for the CI concentration-index forest.

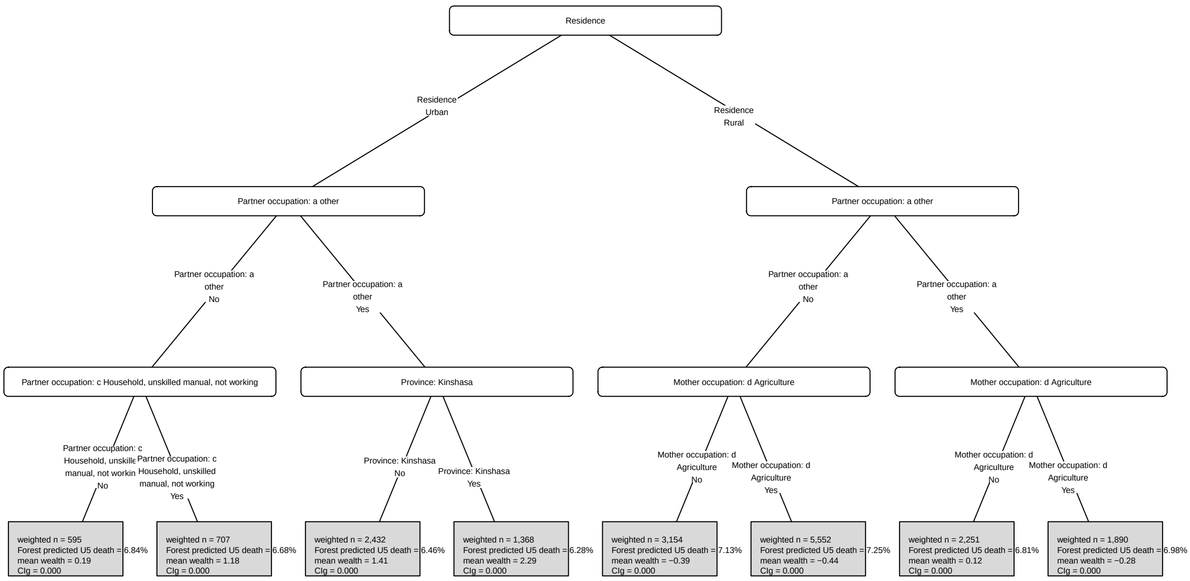

Figure 19: Surrogate tree for the CIg concentration-index forest.

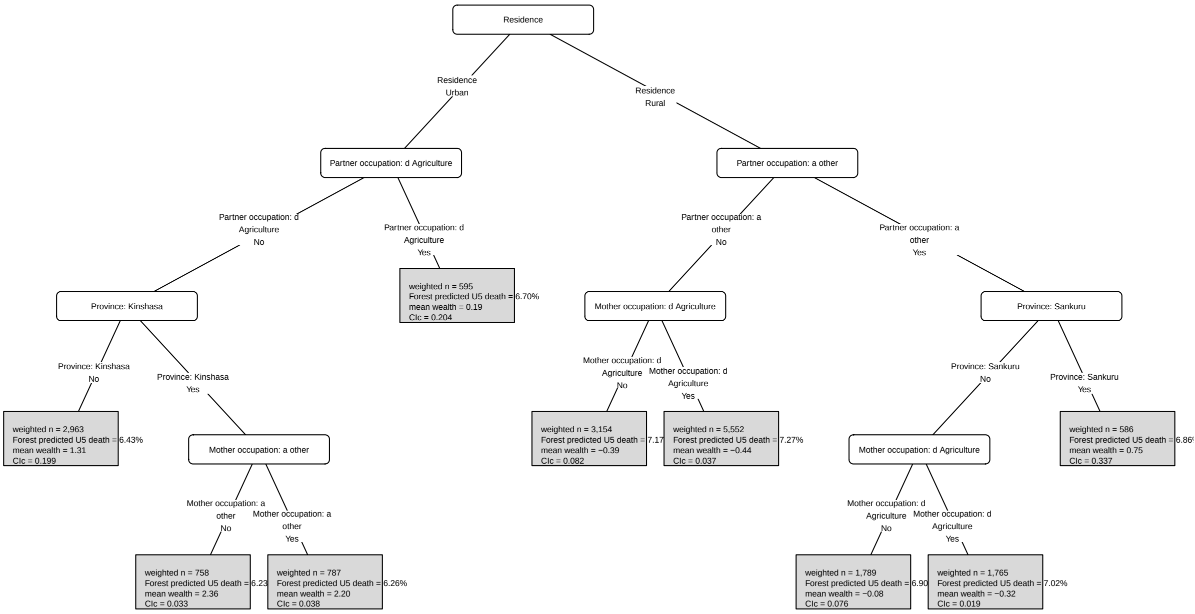

Figure 20: Surrogate tree for the CIc concentration-index forest.

2.1.5 Appendix R5: Implementation Status

Implemented in the report: selected concentration-index tree and selected forest/surrogate-tree variable-importance plots in the results section; tree and concentration-index forest best-control tables, selected-setting validation summaries, selection metrics, validation and complexity plots, and variable-importance plots by criterion in the appendix; ranger cross-validation and SHAP summary outputs in the appendix.

To implement for the final analysis run: rerun the generator with `RUN_CV=true` and `RUN_FOREST_CV=true` when the full cross-validation objects should be recomputed from scratch rather than read from previously generated results.